

Where Major Theorems Live in Proof Space

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A controlled map against LeanDojo Benchmark 4

Abstract

We locate 18,852 Lean declarations from nine repositories containing major recent formalizations in embedding spaces anchored by 10,000 frozen LeanDojo theorem–proof pairs. The repositories are peripheral: their raw mean statement-density deficit is 0.0945 cosine points, and 62.7% of declarations fall below LeanDojo’s own fifth density percentile. A sequence of controls changes the interpretation. Length and coarse domain, exact toolchain date, surface style, source architecture, dependency depth, and reusable compression do not remove the shift. Bound-name and conservatively resolved project-global vocabulary ablations jointly reduce the exact-snapshot deficit by 46.3%, from 0.0600 to 0.0322, but a significant repository-level residual remains. Most importantly, 34 named headline theorems are not more isolated than same-repository, same-domain, architecture-matched declarations. Thus the observed rarity belongs primarily to repository dialect and mathematical ecology, not specifically to celebrated endpoints. Embedding rarity alone is therefore not evidence of theorem-level creativity.

Corpus and estimand. The fixed reference set R contains 10,000 seed-0 LeanDojo declarations. The frontier corpus contains OpenAI’s Ten Proofs and Navier–Stokes/Euler projects, FLT, PFR, PNT+, Formalized Formal Logic, Jordan–Schönflies, Liquid Tensor, and Sphere Eversion. Large repositories contribute a seed-0 sample of about 2,500 declarations plus all identified headline theorems; smaller projects are retained in full. There are 18,852 declarations and 36 named endpoints, of which 34 in eight Lean 4 projects enter the controlled headline analysis. For unit-normalized statement embedding $z_t \in \mathbb{R}^{1024}$, define reference-neighborhood density

$$\rho_k(t) = \frac{1}{k} \sum_{r \in N_k(t)} \langle z_t, z_r \rangle, \quad N_k(t) = \arg \operatorname{topk}_{r \in R} \langle z_t, z_r \rangle.$$

Low ρ_k means that t is locally rare relative to the frozen reference. For each frontier declaration t , let $m(t)$ be an exact-snapshot Mathlib control matched on frozen raw semantic cluster and statement length. Its positive isolation residual is

$$A(t) = \rho_k(m(t)) - \rho_k(t). \quad (1)$$

Project means, rather than thousands of declarations, are the replication units for reported intervals. For an ablated representation a , define attenuation of a negative matched gap Δ by

$$\eta_a = 1 - \frac{|\Delta_a|}{|\Delta_{\text{raw}}|}. \quad (2)$$

Initial map. In the original frozen map at $k = 10$, mean frontier statement density is 0.6717 versus 0.7662 within LeanDojo; proof density is 0.7030 versus 0.7971. Accordingly, 62.7% of frontier statements and 60.7% of frontier proofs lie below the corresponding LeanDojo fifth-percentile thresholds. Semantic neighbors are still useful: statements selected as neighbors have proofs 0.0565 cosine points closer to the frontier proof than random reference proofs. The corpora are peripheral, not detached. Figure 1 displays the same project clouds in statement and proof views.

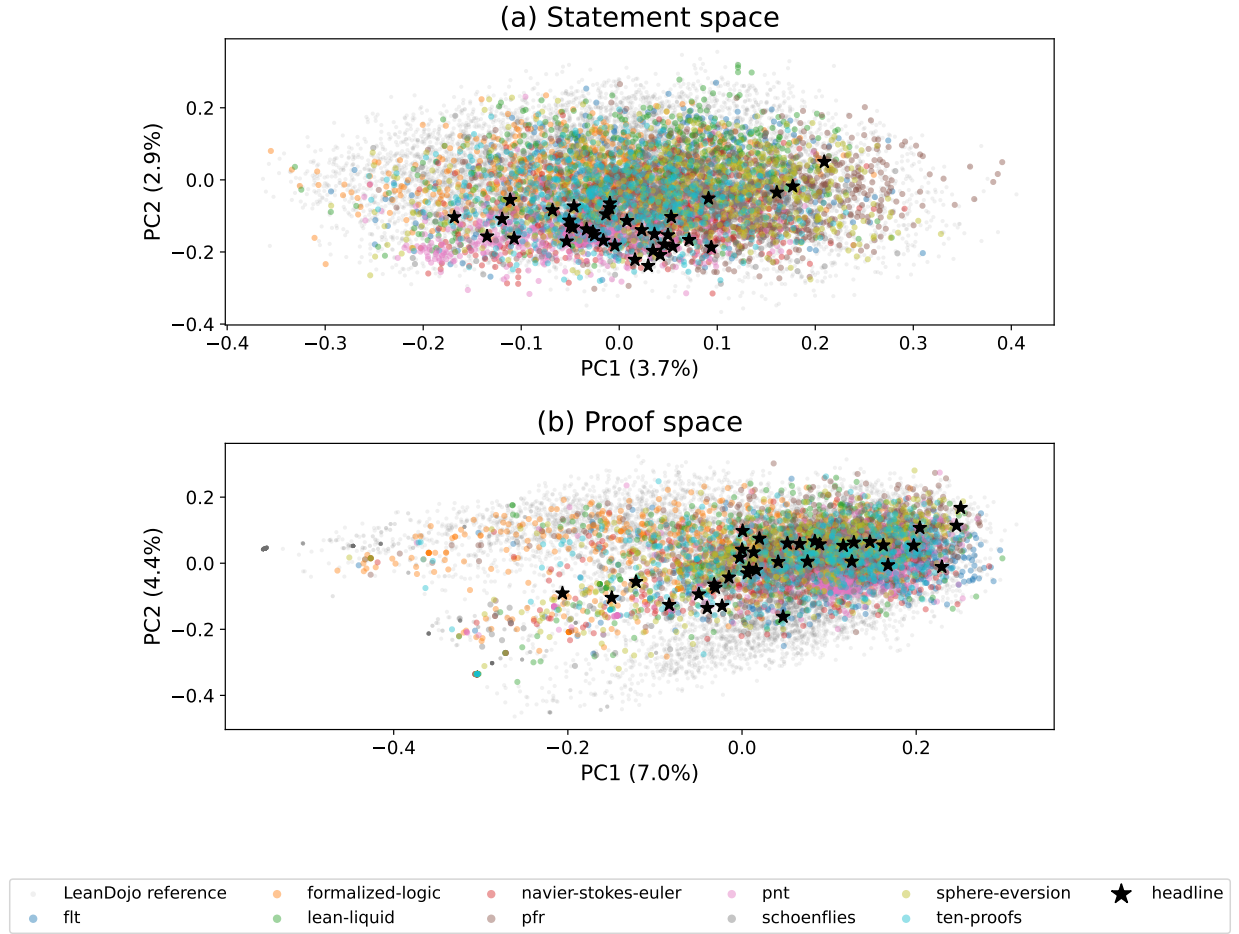


Figure 1: Statement (top) and proof (bottom) embeddings in separate 2D PCA bases fitted only on the corresponding LeanDojo reference (gray). Colors identify repositories and stars mark headline declarations. The displaced clouds establish repository-level distribution shift; they do not show that headline stars are exceptional within their own clouds. Statistics use the full 1,024-dimensional vectors, not these projections.

What explains the displacement? Table 1 summarizes the successive tests. Length plus frozen coarse semantic cluster explains about 22% of the initial shift. Replacing the old reference comparison with 21,000 declarations from the projects’ exact pinned Mathlib snapshots leaves a raw project-balanced gap of -0.0600 . Removing headers does not reduce it. Alpha-normalizing local bound names reduces it to -0.0452 . Static source architecture is not a robust explanation: a linear adjustment gives -0.0338 , whereas a cross-fitted random forest gives -0.0445 . The conclusion depends on model specification rather than revealing a stable mechanism.

Test	Effect	95% project interval	Interpretation
Raw exact-snapshot match	-0.0600	$[-0.0791, -0.0409]$	Recency and toolchain drift do not remove the gap.
Headers removed	-0.0625	$[-0.0840, -0.0410]$	Declaration headers are not the cause.
Bound names also removed	-0.0452	$[-0.0685, -0.0220]$	Local naming explains 24.6% of the raw gap.
Static architecture, linear	-0.0338	$[-0.0596, -0.0080]$	Suggests attenuation, but is model-sensitive.
Static architecture, nonlinear	-0.0445	$[-0.0656, -0.0235]$	Only 1.5% attenuation; architecture is not robust.
Project globals anonymized	-0.0322	$[-0.0508, -0.0136]$	28.8% beyond bound names; 46.3% total attenuation.

Table 1: Matched frontier-minus-control density effects. Negative values mean frontier declarations remain less dense. The ablations identify sensitivity to representation, not causal shares, and their percentages should not be added except along the stated nested raw $\rightarrow$ bound-name $\rightarrow$ project-vocabulary sequence.

The vocabulary test starts from alpha-normalized statements and replaces lexically resolved project-defined global constants by statement-local placeholders $g_0, g_1, \dots$, preserving repeated-symbol identity and Mathlib vocabulary. Resolution is deliberately conservative: exact qualified names are replaced, while ambiguous unqualified names are retained. It changes 10,269 of 18,852 statements and 22,527 symbol occurrences. The remaining -0.0322 gap is significant, so spelling and repository API alone are insufficient. Figure 2(a) shows strong heterogeneity: Formalized Logic and Sphere Eversion approach the reference, whereas PNT+, Navier–Stokes, Ten Proofs, and Schönflies remain displaced.

Dependency geometry. Let $G = (V, E)$ be a source-static dependency graph with $t \rightarrow u$ when the body of t lexically names declaration u . The graph contains 825,962 declarations and 5,009,790 resolved edges. Define the prerequisite cone and synthesis depth

$$P(t) = \{u : t \rightsquigarrow u\}, \quad D(t) = \max_{u \in P(t)} \text{dist}_G(t, u). \quad (3)$$

For prerequisite u , with direct downstream reuse $r(u)$, body length $|p_u|$, and name length $|n_u|$, define a source-level shortcut saving

$$S(u) = \max\{0, (r(u) - 1)|p_u| - r(u)|n_u|\}, \quad C(t) = \frac{\sum_{u \in P(t)} S(u)}{\sum_{u \in P(t)} S(u) + \sum_{u \in P(t)} |p_u|}. \quad (4)$$

Excluding macro-obscured FLT, headline cones are an estimated 2.25 dependency steps deeper than matched declarations (interval $[0.01, 4.48]$), but their compression-fraction lift is -0.026 ($[-0.098, 0.046]$). More importantly, across 16,350 declarations the controlled association between one standard deviation of $\log(1 + D)$ and isolation residual is only 0.0014 ($[-0.0019, 0.0047]$). Depth plus compression add 0.32% five-fold out-of-sample explained variance ($[-0.08\%, 0.73\%]$). Deeper synthesis may describe some endpoints, but it does not explain the semantic displacement.

Are the major theorems themselves rare? This is the decisive distinction. For headline h , let $M_P(h)$ contain up to 20 non-headline declarations from the same repository and raw semantic cluster, nearest in statement/proof length, file position and size, scope, binders, imports, and local-reference features. Define

$$H(h) = \frac{1}{|M_P(h)|} \sum_{c \in M_P(h)} \rho_k(c) - \rho_k(h). \quad (5)$$

Positive H means that a headline is more isolated than comparable local work. Across 34 modern headlines, the project-balanced mean is -0.0060 ($[-0.0252, 0.0133]$). After excluding 5,492 sampled declarations in any headline prerequisite cone, it is -0.0020 ($[-0.0235, 0.0195]$). Excluding FLT’s macro-wrapped endpoints gives 0.0049 ($[-0.0116, 0.0215]$). None differs from zero. Most exact-cluster matches use 20 controls; one sparse PFR cell has two, and matching balance is imperfect, but all specifications give the same null conclusion. Figure 2(b) shows the project effects.

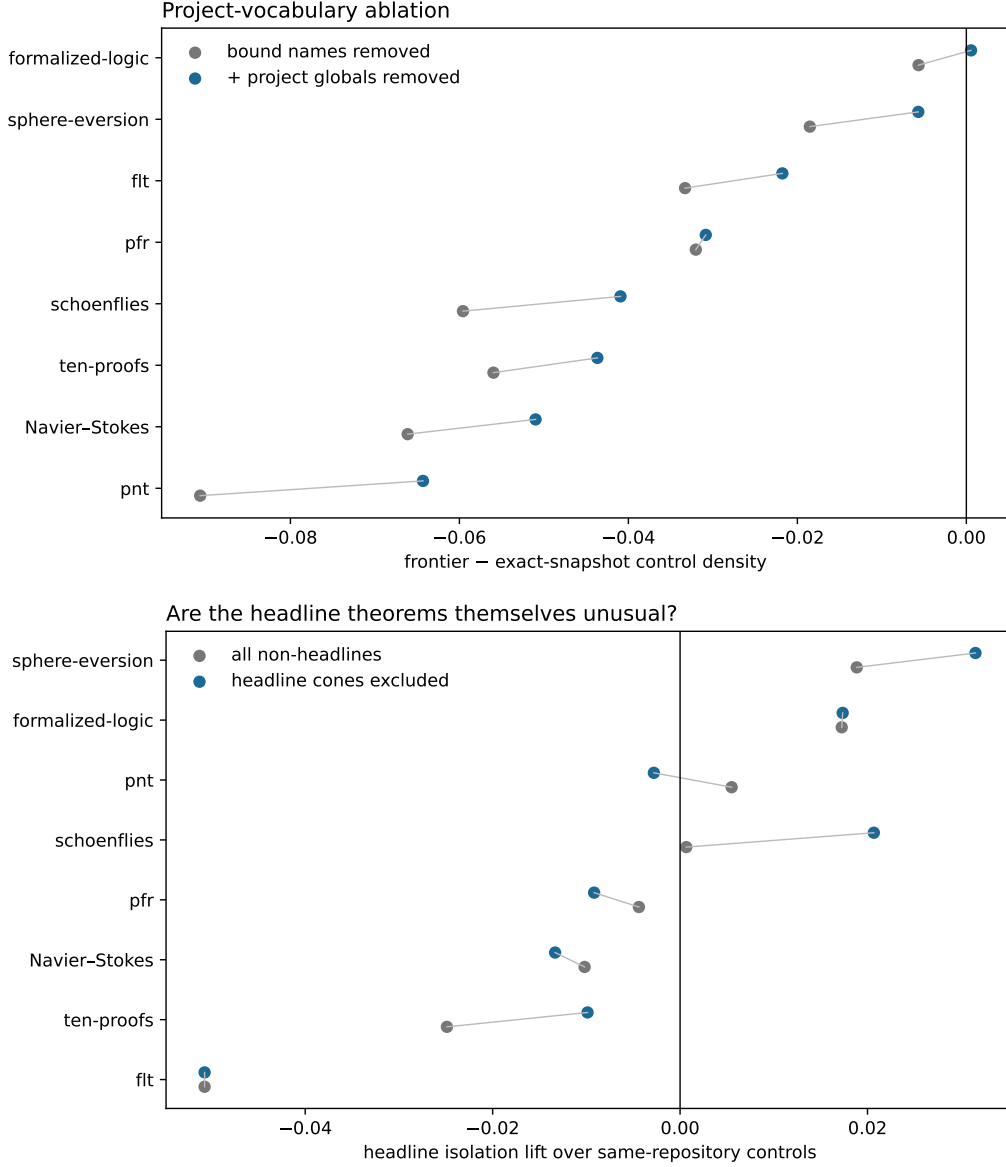


Figure 2: Two tests of what the original project clouds mean. (a) Removing conservatively resolved project-global vocabulary moves every project toward the exact-snapshot reference, but several gaps remain. (b) Within those repositories, named headline theorems are not consistently more isolated than same-domain architecture-matched declarations; positive values would indicate headline-specific rarity.

Creativity formalism. The motivating triad is

$$\mathcal{K}(t) = \underbrace{\mathcal{S}(t)}_{\text{structure preservation}} \times \underbrace{\mathcal{G}(t)}_{\text{compression gain}} \times \underbrace{\mathcal{R}(t)}_{\text{low prior probability}}. \quad (6)$$

The product notation emphasizes that rarity alone is insufficient. For a Lean proof, type checking supplies strong evidence for $\mathcal{S}$. Embedding isolation is at best a proxy for $\mathcal{R}$ under the explicitly chosen LeanDojo prior. Equations (3)–(4) test limited source-level proxies for $\mathcal{G}$; they do not measure conceptual explanatory compression.

The experiments establish repository-level rarity $\mathcal{R}(\text{project} \mid \text{LeanDojo})$, partly induced by vocabulary. They do *not* establish conditional headline rarity $\mathcal{R}(h \mid \text{project})$, because (5) is null, nor do they establish compression gain. Consequently the current map cannot support a claim of theorem-level creativity. A suitable next target is explanatory compression: whether a new abstraction shortens many derivations, generates unusually many nontrivial consequences, or joins previously weakly connected mathematical modules while preserving correctness.

Conclusion and limits. Where do major theorems live? Repositories containing them live near the edge of the frozen LeanDojo space. The named endpoints themselves do not occupy a detectably more peripheral region than comparable declarations in those same repositories. Roughly 46% of the controlled displacement is sensitive to local and project-global naming; the residual is best described as repository-level mathematical-content and representation shift, not “profoundness.”

The remaining limitations are substantive. Embeddings measure textual representation rather than mathematical equivalence. Global-symbol anonymization and dependency resolution are lexical, FLT hides structure behind proof macros, exact-cluster control cells can be sparse, and eight projects are few independent replication units. Elaborated dependencies and a direct measure of consequence-level compression are required before revisiting the creativity hypothesis.

Reproducibility. Frozen commits, inputs, sampling, controls, code, figures, and machine-readable results are retained under `experiments/frontier-*`. The final vocabulary ablation re-embedded only frontier statements, reused prior reference/control vectors, and cost an estimated \$0.162. All experiments remained below the precommitted \$15 AWS cap.